



A.1 Kinematics

Space, time and motion · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Kinematics describes motion using displacement, velocity and acceleration without reference to forces. For uniformly accelerated motion in one dimension, the equations $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$, and $s = \frac{1}{2}(u+v)t$ connect the variables, where u is initial velocity, v is final velocity, a is constant acceleration, s is displacement and t is time. Projectile motion is analysed by treating horizontal (constant velocity) and vertical (constant acceleration g) components independently, since they are perpendicular and independent.

Worked example: follow the reasoning

A ball is thrown horizontally at $12 \text{ m}\cdot\text{s}^{-1}$ from a cliff of height 45 m. Find the time to reach the ground and the horizontal distance travelled.

Vertical: $s = \frac{1}{2}gt^2 \rightarrow 45 = \frac{1}{2}(9.81)t^2 \rightarrow t^2 = 9.17 \rightarrow t = 3.03 \text{ s}$.

Horizontal: $x = v_h \cdot t = 12 \times 3.03 = 36.4 \text{ m}$.

Answer: $t \approx 3.03 \text{ s}$, $x \approx 36.4 \text{ m}$.

Quick check · multiple choice

A car accelerates uniformly from rest to $20 \text{ m}\cdot\text{s}^{-1}$ in 5.0 s. What distance does it travel?

- A 20 m
- B 50 m
- C 100 m
- D 200 m

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A stone is launched from ground level with initial speed $25 \text{ m}\cdot\text{s}^{-1}$ at 40° above the horizontal. Air resistance is negligible and $g = 9.81 \text{ m}\cdot\text{s}^{-2}$. [6]
- (a) Calculate the initial vertical and horizontal components of velocity.
 - (b) Calculate the maximum height reached.
 - (c) Calculate the total time of flight and the horizontal range.

Final answer: _____



Answers and explanation

Question 1 [6]

(a) $v_x = 25\cos 40^\circ = 19.2 \text{ m}\cdot\text{s}^{-1}$; $v_y = 25\sin 40^\circ = 16.1 \text{ m}\cdot\text{s}^{-1}$.

(b) Max height: $v_y^2 = 2gh \rightarrow h = 16.1^2 / (2 \times 9.81) = 13.2 \text{ m}$.

(c) Time up = $v_y/g = 1.64 \text{ s}$, total time = 3.28 s . Range = $v_x \times t = 19.2 \times 3.28 = 63.0 \text{ m}$.

Quick check answer

B. $s = \frac{1}{2}(u+v)t = \frac{1}{2}(0+20)(5.0) = 50 \text{ m}$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.